

Certificate of Analysis

AAV drug substance release

PRODUCT AND TRACEABILITY

Product / molecule	OBX-201 (recombinant AAV9)	Material	Drug substance
Drug substance lot		Manufacturing date	
CoA issue date		Storage condition	Store at or below minus 60 C
Serotype / route	AAV9, intravenous (systemic)	Specification ID / version	SPEC-DS-OBX201 v1.0
Plasmid lots (RepCap / Helper / Transgene)	PL-REPCAP-0417 / PL-HELP-0417 / PL-TRANS-0417	Working cell bank lot	WCB-293-018
Batch record reference	MBR-USP-201, MBR-DSP-201	Backing worksheets	WS-AEX-2605X, WS-VG-2605X

RELEASE TEST PANEL

Test	Method reference	Acceptance criterion	Result	Pass / Fail
Identity				
Vector genome identity (transgene)	qPCR/ddPCR ITR and transgene, SOP-3510	Conforms to reference sequence		
Capsid and serotype identity	LC-MS intact mass VP1/VP2/VP3, SOP-3512	VP masses within 5 Da; AAV9 confirmed		
Strength				
Vector genome titer	ddPCR, SOP-3220 (see WS-VG-2605X)	1.6 to 2.4 x 10 ¹³ vg/mL		
Capsid particle titer	Capsid ELISA, SOP-3520	Report		
Purity and product-related impurities				
Percent full capsids	AEX-HPLC, SOP-3210 (see WS-AEX-2605X)	Not less than 80.0 percent full		
Aggregates and percent monomer	SEC-MALS, SOP-3530	Not less than 95 percent monomer		
Process-related impurities				
Residual host cell DNA	qPCR, SOP-3540	Less than 10 ng per dose, under 200 bp		

Residual plasmid DNA	qPCR, SOP-3542	Less than 100 ng per 1e12 vg		
Residual host cell protein	ELISA, SOP-3544	Not more than 100 ng/mL		
Residual Benzonase	ELISA, SOP-3546	Not more than 1 ng/mL		
Potency				
Infectious titer	TCID50 cell-based, SOP-3550	Report IU/mL		
Vector genome to infectious unit ratio	Calculated (VG titer / infectious titer)	Not more than 100		
Relative potency	Cell-based transgene expression, SOP-3552	50 to 150 percent of reference		
Safety				
Sterility	USP <71>, SOP-3560	No growth		
Endotoxin	USP <85> LAL, SOP-3562	Less than 5 EU/kg/hr		
Mycoplasma	USP <63>, SOP-3564	Negative		
Replication-competent AAV	Cell-based amplification with qPCR, SOP-3566	Less than 1 per 1e10 vg		
General				
Appearance	Visual inspection, SOP-1805	Clear to slightly opalescent, colorless		
pH	USP <791>, SOP-3570	7.4 plus or minus 0.4		
Osmolality	USP <785>, SOP-3572	280 to 400 mOsm/kg		

Lines annotated "see WS-..." are backed by the referenced analytical worksheet. Units differ by attribute (vg/mL, capsids/mL, ng per dose, EU/kg, percent full, VG:IU).

OVERALL COMPLIANCE STATEMENT

State conformance to specification SPEC-DS-OBX201 v1.0 and to GMP, the replication-competent AAV and adventitious-agent status, and the disposition (Released, Rejected, or Quarantine / Under Investigation).

QA AUTHORIZATION

Role	Name	Signature	Date
Reviewed by (technical review)			
Approved by (Quality)			

Assurance)			
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The QA approval and date are the release act. This certificate is not valid unless approved by Quality Assurance.